NEET Important Chapter - Thermal Properties of Matter

Important Concept of Thermal Properties of Matter for NEET

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This chapter deals with the basics of thermodynamics. It starts with very basic concepts like temperature and heat. In this chapter, we study temperature, heat, thermal expansion of the material, and different modes of heat transfer.

When we read thermal properties of matter the first thing that strikes our mind is, what are the thermal properties of matter? To answer this question we need to go through the chapter but we can brief it by saying that when the temperature is applied to any material, then how it will act is known as the thermal property of that material.

Now, let's move on to the important concepts and formulae related to the NEET exam along with a few solved examples from the chapter on thermal properties of matter.

Some Important Topics of Thermal Properties of Matter for NEET

- Temperature and Heat
- Thermal Expansion
- Thermal Stress
- Calorimetry
- Conduction
- Convection
- Radiation
- Wien's Displacement Law
- Stefan-Boltzmann Law
- Newton's Law of Cooling

Some Important Concept of Thermal Properties of Matter for NEET

Name of the concept	**Key points of the concepts**
1. Temperature	The term "temperature" refers to a physical quantity that describes how hot or cold something is. Temperature is a manifestation of thermal energy, which is present in all matter and is the source of heat, a flow of energy when one body comes into contact with another that is colder or hotter.

2. Heat	When two bodies are at different temperatures then energy transfer takes place between these two bodies, it is known as heat or heat flow. Two bodies at different temperatures are brought together then energy is transferred from the hotter body to the colder. So we can say Heat can not be defined for a single body. It needs two references to define it.
3. Thermal Expansion & Thermal Stress	• The tendency of matter to change shape, area, volume, and density in response to temperature changes is known as thermal expansion. • When a material goes through a rapid change in temperature, its surface and the inner temperature difference are created and it causes a fracture in the material. It is known as thermal stress.
4. Calorimetry	Calorimetry is a concept to measure the state variable of a body so that heat transfer can be calculated using the initial and final state variable values.

5. Modes of Heat Transfer	There are three modes of heat transfer- • **Conduction**: Thermal conduction is the transfer of internal energy by microscopic collisions of particles and the movement of electrons within a body. Medium is required to transfer heat via conduction. • **Convection**: Thermal convection is the process of heat transfer by the bulk movement of molecules within fluids such as gases and liquids. Medium is required to transfer heat via convection. • **Radiation**: Due to thermal motion of particles a electromagnetic radiation is generated which is known as thermal radiation. In nature all matter with a temperature greater than absolute zero or we can say (Zero Kelvin) emits thermal radiation.
6. Wien's Displacement Law	According to the Wien's displacement law the black-body radiation curve for different temperatures will peak at different wavelengths that are inversely proportional to the temperature.

7. Stefan-Boltzmann Law	The total radiant heat power emitted from a surface is proportional to the fourth power of the absolute temperature, according to the Stefan-Boltzmann law.
8. Newton's Law of Cooling	According to Newton's law of cooling, the rate at cooling is proportional to the difference in temperature between the object and the object's surroundings.

List of Some Important formulas of Thermal Properties of Matter

S.No.	Name of the Concept	Formula
1.	**Linear Expansion**	
2.	**Spherical Expansion**	
3.	**Volumetric Expansion**	
4.	**Temperature of Mixture**	
5.	**Newton's Law of Cooling**	
6.	**Approximation Method**	
7.	**Stefan-Boltzman Law**	Where, Q= Rate of heat radiation A = Surface Area = Stefan- Boltzman Constant T = absolute temperature
8.	**Wein's displacement Law**	Where, = Wavelength corresponding to maximum energy b = Wein's displacement constant T = Temperature

Solved Examples of Thermal Properties of Matter

1. **A circular hole of diameter 2.00 cm is made in an aluminium plate at . What will be the diameter at ? (Linear expansion for aluminium =)**

Sol:

Given that,

Diameter of hole = cm

Initial Temperature =

Final Temperature =

Coefficient of linear expansion = =

To find: Diameter of hole at

Formula used:

Using the above formula,

In order to solve these kinds of problems we need the standard formula for thermal expansion. By putting values in the formula we can get the desired answer.

2. **A steel tape 1m long is correctly calibrated for a temperature of . The length of a steel rod measured by this tape is found to be cm on a hot day when the temperature is . What is the actual length of the steel rod on that day? What is the length of the same steel rod on a day when the temperature is ? (Coefficient of Linear Expansion of steel =)**

Sol:

Given that,

Length of steel rod at temperature ,(=), =

Length of steel rod at temperature ,(=   ),   =

Coefficient of linear expansion =   =

To find: Actual Length of rod

Formula used:

Using the above formula,

First let's assume that   is the actual length of a steel rod.   is the length of the tape when temperature is   .

Now we have a length of tape on the day when the temperature is   . So actual length of the rod on that day-

Now length of rod when temperature is   ,

Key point: In order to solve these kinds of problems we need the standard formula for thermal expansion. By putting values in the formula we can get the desired answer.

Previous Year Questions of Thermal Properties of Matter from NEET Paper

1. **The copper rod of and an aluminium rod of unknown length have an equal increase in their lengths independent of the increase in temperature. The length of the aluminium rod is (and) ( )**

a.

b.

c.

d.

Sol:

Given that,

Length of copper rod =

Coefficient of linear thermal expansion of Copper=

Coefficient of linear thermal expansion of Aluminium=

To Find: Length of aluminium rod

Concept and Formula Used: Due to change in temperature, the thermal strain produced in the rod of length L is given by -

Where, L = Original length of rod,

= Coefficient of linear expansion of rod,

= Change in length of rod.

As per the question, increases in the lengths are equal and it is independent of temperature change. Hence,

Hence, the length of the Aluminium rod is .

Hence, option (C) is correct.

Trick: In order to solve these problems, we have to search for the concept which is applicable to the question. Like in this question linear thermal expansion was used to solve the question.

2. **The cup of coffee cools from to in minutes, when the room temperature is . The time taken by a similar cup of coffee to cool from to at a room temperature same at is ( )**

a.

b.

c.

d.

Sol:

Given that,

the initial temperature of a cup of coffee is, =

the final temperature of a cup of coffee is, =

Time taken to drop the temperature=

Room temperature= =

To Find: Time taken by a similar cup of coffee to reduce temperature from to .

Formula Used: To solve this question we will use Newton's law of cooling. According to Newton's law of cooling-

If we use Newton's law of rate of cooling for first temperature drop,

Similarly, for second temperature drop-

By putting value of K

Hence, option (B) is correct.

Trick: In order to solve this problem we must have an understanding of Newton's law of cooling.

Practice Questions For Thermal Properties of Matter

1. Hot water cools from to in the first minutes and to in the next minutes. What is the temperature of the surroundings?
(Ans.)

2. 2 kg of ice at is mixed with 5 kg of water at in an insulating vessel having a negligible heat capacity. Calculate the final mass of water (in kg) remaining in the container. It is given that the specific heats of water and ice are and while the latent heat of fusion of ice is .
(Ans. 6)

Conclusion

In this chapter we discussed the important topic of the NEET exam. Thermal Properties of Materials are very important and basic of whole thermodynamics. We discussed some important concepts like Temperature and Heat, Heat Transfer, Rate of Cooling, and Different modes of heat transfer.